Data Engineering Project Workflow

Instruction Document

1. Project Setup:

- Created project folder
- Created virtual environment (venv) inside the VS code
- Installed required libraries
- Generated requirements.txt file

2. Web Scraping:

- Built scraper using Playwright in VS Code (file name→ scraper.py)
- Scraped data from IndiaMART website (single page)
- Extracted product details and stored data in JSON format
- Output file name → indiamart_data.json

3. Data Processing:

- Launched Jupyter Notebook using Anaconda Prompt
- Imported required Python libraries (pandas, matplotlib, etc.)
- Loaded JSON dataset into DataFrame

4. Data Cleaning:

- Removed currency symbols and converted price to numeric
- Handled missing values (filled using mean)
- Ensured data consistency

5. Data Preparation:

- Extracted city from location column
- Created new features for analysis

6. Data Visualization:

- Created histogram for price distribution
- Analyzed price by location and supplier
- Generated keyword frequency charts
- Built correlation analysis (price vs rating)

7. Insights & Hypothesis:

- Identified pricing trends and supplier patterns
- Observed regional concentration of suppliers
- Found weak correlation between price and rating
- Formulated hypothesis on supplier trust based on ratings